LINGVESSA

Technical Design

Implementation architecture and contracts / v1.7

Concrete design for the bilingual production-oriented reference implementation: modular monolith, PostgreSQL, versioned learning/content/media contracts, two-level mastery, pronunciation UX, AI for Learning + Administration, provider routing, security, delivery and executable acceptance.

CORE ENGLISH + ČEŠTINA / ONE SHARED ENGINE
LEARNING STATE POSTGRESQL + IMMUTABLE EVIDENCE / PUBLICATIONS
MASTERY 6 GLOBAL SKILLS + KNOWLEDGE-ITEM FACETS
AI OPTIONAL / PROVIDER-AGNOSTIC / FREE-LOCAL-FIRST / PAID BUDGET 0
DEV DEFAULT GITHUB PRIVATE + NORTHFLANK SANDBOX + CLOUDFLARE R2

Status: OWNER-LOCKED IMPLEMENTATION BASELINE
Updated: 9 September 2026| Replaces Technical Design v1.6

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-TECH-DESIGN
Версія / дата	1.7 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Замінює	Technical Design v1.6; previous versions remain historical.
Governing documents	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
Архітектура	TypeScript modular monolith + workers; PostgreSQL source of truth; S3-compatible object storage; provider-neutral ports.
Стартові мови	English en-GB default/en-US variant; Čeština cs-CZ; UI/explanations uk-UA.
AI invariant	Core course, attempts, mastery, review and published media remain functional with every AI provider disabled.
Development default	Private Git repository; GitHub Private + Northflank Developer Sandbox + Cloudflare R2 as current low-cost development choice; not production lock-in.

Що змінено у v1.7

Зміна	Результат
Baseline sync	Dependencies updated to Formal v1.7, Technical v1.7, Curriculum A1 v1.6, Reference v1.6 and AI Integration v1.4.
AI Layer	Added AI Gateway/Orchestrator, Policy Engine, Provider Router, Prompt Registry, Grounding Builder, Output Validator, Usage Meter and audit contracts.
AI learning	Concrete contracts for explain differently, bounded hints, extra practice, writing feedback and text role-play; non-AI fallback is mandatory.
AI administration	Manual / AI Assist / AI Auto permissions, low-risk allowlist, proposal/diff workflow and prohibition on direct publish/delete/role/billing actions.
Cost protection	Paid inference disabled by default, monthly paid AI budget = 0, free/local-first routing and hard pre-call budget gate.
Data model	Added provider/model/policy/prompt/run/session/usage entities while preserving immutable learning evidence and two-level mastery.
Lesson schema	Optional ai_extensions added as non-authoritative capability references; schema remains complete when field absent/disabled.
Reference suite	Reference Module Design v1.6 32-scenario regression suite becomes mandatory downstream fixture; AI-off path is a release gate.
Deployment	Development default documented without binding production architecture to Northflank/R2/GitHub vendor APIs.
Prototype	No disposable one-folder prototype stage; implementation begins with production-oriented bilingual reference slice.

Precedence note
Якщо цей design суперечить governing product/technical/AI document, має пріоритет пізніше затверджене governing requirement. Implementation convenience не змінює product scope без ADR/spec revision.

Зміст

1. Призначення, precedence і статус
2. Fixed architecture and ADR set
3. Runtime topology and synchronous boundary
4. Technology, repository and module ownership
5. Environments, configuration and development deployment
6. Data conventions and non-negotiable invariants
7. Canonical relational model v1.7
7.1 Critical keys and database guards
8. Publication and content-version contract
9. Learning, mastery, review and outbox DDL contract
9.1 Attempts and evidence
9.2 Two-level mastery and review
10. AI domain model and persistence contract
11. Authoring tree, release package and localization
12. Lesson Content Schema v1.7
13. Activity registry and type contracts
14. State machines and completion contracts
14.1 Scored activity
14.2 Pronunciation binary UX with safe fallback
14.3 AI conversation/help states
15. Answer evaluation and normalization
16. Attempt command, transaction and idempotency
17. Evidence and two-level mastery model
18. Mistake remediation, reset and Review Engine
19. AI for Learning runtime contract
20. Media, audio, voices and speech
21. AI orchestration, provider routing and cost controls
22. AI Administration: Manual / Assist / Auto
23. Jobs, queues, outbox and callbacks
24. API conventions and surfaces

24.1 Learner and AI APIs
24.2 Admin/content/AI APIs
25. Web/PWA and LessonPlayer lifecycle
26. Admin/CMS frontend and AI controls
27. Authentication, authorization and application security
28. Privacy, consent and data lifecycle
29. Performance, reliability and recovery
30. Observability and domain-event contract
31. CI/CD, migrations, evals and release gates
32. Reference modules as executable fixtures
32.1 Technical acceptance scenarios
33. Definition of Done
34. Implementation sequence
35. Deferred choices and fixed baseline
36. Sources and change control

ЧАСТИНА A. РІШЕННЯ, АРХІТЕКТУРА ТА МЕЖІ

1. Призначення, precedence і статус

Technical Design v1.7 переводить затверджені продуктні та технічні вимоги Lingvessa у конкретні межі модулів, структури даних, state machines, transaction boundaries, provider contracts, API payloads, AI capability policies, release gates та executable acceptance. Документ не є доказом реалізації: acceptance настає лише після коду, tests, QA evidence і owner approval.

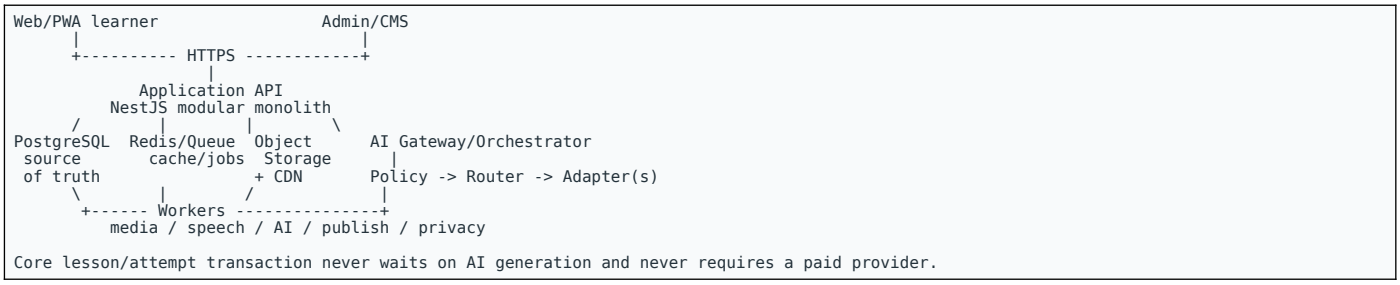
- Одна платформа обслуговує English та Čeština через спільні engines; language-specific differences задаються configuration/strategy, не fork application.
- Core learning path не залежить від AI, live media generation або доступності зовнішнього speech provider.
- Published content і assets immutable/versioned; attempts/evidence immutable; rollback не переписує історію.
- AI може допомагати, але не є source of truth для score, mastery, due_at, publication state або privileged action.
- All code, schemas, migrations, scripts and manifests are version-controlled in a private repository; media bytes and secrets are not committed as normal source artifacts.

ID	Вимога	Пріоритет	Критерій перевірки
TD-GOV-001	Every MUST requirement in governing specs has implementation trace to schema/code/test/ADR.	MUST	Release checklist has no orphan MUST requirement.
TD-GOV-002	AI optionality is architectural, not UI-only.	MUST	All providers disabled still passes core English/Czech reference E2E.
TD-GOV-003	Behavior-changing design update increments controlled design version.	MUST	Change log, migration/compatibility note and affected tests exist.
TD-GOV-004	Vendor choice cannot leak into domain/content identity.	MUST	Provider replacement contract tests pass; no provider URL in canonical content.

2. Fixed architecture and ADR set

ADR / decision	Fixed design	Reason / invariant
ADR-D01	Modular monolith API + separate workers	Fast iteration, enforceable boundaries; no premature microservices.
ADR-D02	PostgreSQL authoritative domain state	Progress, attempts, mastery, review, publications, AI policy/audit survive cache/provider failures.
ADR-D03	Content-driven reusable Lesson Engine	Educational text/order/answers are versioned data, not hardcoded React conditionals.
ADR-D04	Two-level mastery	Six global skill projections + knowledge-item facets; one percentage never replaces granular evidence.
ADR-D05	Server-authoritative answer/evidence	Client cannot claim correct/mastery/due_at; AI output cannot bypass evaluator.
ADR-D06	Binary pronunciation UX only when usable	speech_correct/speech_incorrect for sufficient confidence; speech_unavailable is neutral/non-blocking.
ADR-D07	Pre-generated versioned media	Normal lesson resolves stable asset_id + asset_version; live generation is not required.
ADR-D08	Provider-neutral AI/TTS/STT/image adapters	Cloud/free/local/paid models remain replaceable; SDK types stay in infrastructure.
ADR-D09	AI paid budget = 0 by default	No accidental billing; explicit admin action required before any billable route.
ADR-D10	AI Auto = allowlisted low-risk only	No direct publish, rollback, delete account, role/MFA/security or billing changes.
ADR-D11	Production-oriented reference implementation first	E-A1-M01 + C-A1-M01 validate architecture before bulk A1 content.

3. Runtime topology and synchronous boundary



Component	Owns / responsibility	Must not do
Web/PWA	Learner UX, local unsent queue, media playback/record, AI entry UI, accessible state.	Authoritative scoring/mastery; secrets; direct provider API calls.
Admin/CMS	Structured editing, reviews, AI proposals, jobs, publish/rollback, audit views.	Bypass approvals/policy; expose provider secrets/raw audio.
Application API	AuthZ, use cases, evaluator, transactions, mastery/review projection commands, AI capability gateway.	Long-running provider work inside DB transaction.
PostgreSQL	Identity, content/publication, attempts/evidence, mastery/review, AI policy/run/audit metadata, outbox.	Large media bytes; hidden model reasoning.
Redis/Queue	Cache, rate limits, job queue/leases.	Sole copy of progress/publication/AI audit.
Object storage/CDN	Immutable media bytes; versioned delivery.	Canonical provider URL dependency.
Workers	Media/speech/AI generation, validation, notifications, projections, deletion/export.	Direct publish or privileged mutation outside workflow service.
AI Gateway	Policy, grounding, routing, structured validation, fallback, usage telemetry.	Direct mastery mutation; tool execution without server authorization.
Providers	TTS/STT/image/text capabilities behind adapters.	Receive unnecessary user data or authoritative domain identifiers.

Synchronous boundary
Catalog, lesson manifest, deterministic exercise evaluation, attempt commit, progress, review reads and non-AI remediation stay available when AI/media generation providers are offline. Speech may use provider processing, but failure maps to speech_unavailable rather than blocking the lesson.

4. Technology, repository and module ownership

Layer	Baseline	Boundary
Runtime	Node.js 24 LTS + TypeScript strict + pnpm workspaces	Exact versions pinned in repository/lockfile; LTS only for production.
Web	React + Next.js App Router + PWA + original design tokens	Client boundary receives validated manifests; no secrets.
API	NestJS modular monolith + REST/OpenAPI 3.1.2	/api/v1; Fastify optional after benchmark.
Data	PostgreSQL 18 current minor + Prisma migrations + SQL constraints	Relational ownership; JSONB only for versioned flexible payloads.
Queue/cache	Redis-compatible + BullMQ-compatible adapter as needed	Disposable cache; durable domain data elsewhere.
Media	S3-compatible object storage + CDN + FFmpeg/image tooling	Immutable version/checksum; platform-owned keys.
AI	Internal AI capability contracts + provider adapters	No provider SDK types in domain/content schemas; structured outputs validated.
Testing	Unit/property + Testcontainers + Playwright + a11y/visual + AI evals	English/Czech fixtures + provider doubles + limited sandbox tests.
Delivery	OCI containers, CI/CD, secret manager, IaC-capable config	Dev/staging/prod reproducible; artifact promoted by commit SHA.

/apps/web learner PWA
/apps/admin CMS/admin
/apps/api NestJS HTTP application
/apps/worker queues/media/speech/AI/publication/privacy
/packages/domain framework-free entities/value objects
/packages/application use cases/ports/transactions
/packages/activity-schemas JSON Schema + fixtures + migrations
/packages/lesson-engine state/render contracts
/packages/mastery-review deterministic projections/scheduler
/packages/ai-core capability/policy/request/output contracts
/packages/provider-adapters provider-specific infrastructure: ai/, speech/, tts/, image/, email/, storage/
/packages/api-client generated/validated API client
/packages/ui shared accessible UI primitives
/packages/content-tools lint/import/manifests/preview
/content/english /content/czech /assets/specs /prisma /infra /docs/adr

Module	Owns	May call
identity	users, sessions, roles, consents	audit, notification ports
curriculum/content	languages, knowledge, lessons, content versions, publications	media, AI draft ports, audit
learning	lesson runs, attempts, evidence, mistakes	content read model, mastery/review ports
mastery/review	skill/facet states, schedules, sessions	immutable evidence/content metadata
media/speech	assets, versions, voices, recording/scoring jobs	provider adapters, audit
ai/orchestration	AI policies, routing, prompts, runs, sessions, usage	content/knowledge read models, mastery summary, adapters, audit
collections	personal sets	knowledge/content read refs
admin/workflow	approvals, publish/rollback, privileged command coordination	all modules through application ports
analytics/ops	outbox consumers, projections, metrics	events only; never authoritative writes

ID	Вимога	Пріоритет	Критерій перевірки
TD-MOD-001	Domain packages import no NestJS/Prisma/React/provider SDK.	MUST	Dependency lint passes.
TD-MOD-002	Cross-module write uses declared application port/event, not direct foreign table mutation.	MUST	Architecture tests + code review.
TD-MOD-003	AI provider SDK exists only in infrastructure adapter.	MUST	Forbidden-import rule blocks PR.
TD-MOD-004	AI tool execution re-enters normal application authorization/use-case layer.	MUST	Red-team tool call cannot bypass RBAC/object policy.

5. Environments, configuration and development deployment

Environment	Purpose	Rules
local	Synthetic bilingual fixtures; local DB/queue/storage/adapters.	One-command bootstrap; no production secrets; local AI optional.
preview	Per-PR Web/API preview with ephemeral/synthetic data.	Protected access; auto-expiry; no privileged production integration.
staging	Production-like topology with sandbox/free providers and synthetic/anonymized data.	Migration, E2E, publish/rollback, provider fault, restore and AI evals.
production	Real users/content; managed monitored services.	Least privilege, MFA admin, approved provider/data policy, backups and incident ownership.

Development default
Source: GitHub Private. Current low-cost reference deployment MAY use Northflank Developer Sandbox for Web/API/PostgreSQL and Cloudflare R2 Standard for media. These are configuration/ADR choices, not domain dependencies. Northflank Sandbox is development/testing only; production provider/region remains an explicit ADR.

- Typed environment schema; fail fast on invalid required values.
- Secrets injected at runtime; no provider key in Git, client bundle, logs or content JSON.
- AI configuration includes global enabled, learner/admin enabled; learner AI OFF by default until opt-in/consent, Manual/Assist/Auto, free_only, paid_budget, provider/model allowlist and per-capability quotas.
- Paid AI default is disabled with hard budget 0; no fallback may silently route to billable provider.
- Container/build artifact is immutable and promoted between environments; migrations use expand/migrate/contract.

ID	Вимога	Пріоритет	Критерій перевірки
TD-ENV-001	Config schema is identical across environments; credentials/resources differ.	MUST	Startup/config contract tests.

ID	Вимога	Пріоритет	Критерій перевірки
TD-ENV-002	Secrets never appear in repository/client/build logs/telemetry.	MUST	Secret scan + bundle/log inspection.
TD-ENV-003	Development hosting can be replaced without domain/schema change.	MUST	Container + externalized config portability review.
TD-ENV-004	Paid AI is OFF and budget 0 by default.	MUST	Config test rejects billable route before provider call.
TD-ENV-005	Feature flags may scope language/course/activity/AI capability/provider/cohort.	SHOULD	Flag failure uses documented safe default; never substitutes authorization.

ЧАСТИНА B. ДАНІ, CONTENT VERSIONING ТА LEARNING STATE

6. Data conventions and non-negotiable invariants

Concern	Design
Identity	Opaque UUID; UUIDv7 preferred where supported. Slug/title/order/filename/provider identifier are not identity.
Time	PostgreSQL timestamptz UTC; user timezone IANA; review day boundary computed server-side.
Language	BCP 47: uk-UA, en-GB, en-US, cs-CZ. Target language and UI locale are separate.
Unicode	UTF-8/NFC comparison policy per field. Czech diacritics are never globally stripped.
Scores	Internal estimates 0..1 + model_version; UI percentage is derived presentation.
Mastery	Global SkillProjection is derived; KnowledgeMasteryState is profile+knowledge+facet+model.
AI data	Store final/structured outputs and trace metadata only as policy requires; never hidden chain-of-thought.
JSONB	Versioned content/provider/event payloads only; ownership/status/due_at/joins remain relational.
Audit	Attempts/evidence/publish/audit/outbox append-only/immutable except documented redaction/tombstone.
Language isolation	Every learner state path begins with user + language_profile; cross-language links rejected server/DB side.

Isolation invariant
Switching English <-> Čeština changes active UI context only. It must not mutate the other language progress, skill/facet mastery, due reviews, collections, AI conversation state or course position.

7. Canonical relational model v1.7

Area	Tables / entities	Critical rule
Identity	users, credentials, sessions, roles, user_roles, consents	Ownership server-derived; admin role changes audited.
Profiles	language_profiles, user_settings, notification_preferences	UNIQUE user+language; selected English variety; timezone IANA.
Curriculum	languages, variants, courses, levels, modules, lessons, activities/items	Stable IDs; language lineage consistent.
Knowledge	outcomes, knowledge_items, knowledge_edges, item_knowledge_map	Assessable item maps to skill + relevant facet.
Publication	content_entities, content_versions, publications, entries, approvals, translations	Published snapshot immutable; rollback pointer atomic.
Learning	lesson_runs, activity_attempts, evidence_events, mistake_instances	Attempts/evidence immutable; exact content/evaluator versions retained.
Mastery	skill_mastery_states, knowledge_mastery_states	Separate global skill vs item facet projections.
Review	review_schedules, review_sessions/items	Schedule key includes knowledge+facet+scheduler version.
Collections	collections, entries	Owner + language isolated; deletion does not cascade to knowledge/mastery.
Media	requirements, jobs, assets, asset_versions, voices, characters	Immutable asset versions; provenance required.
AI	ai_provider_profiles, ai_model_profiles, ai_policies, ai_prompt_templates, ai_runs, ai_sessions/messages, ai_usage_buckets	Policy-driven routing; final/structured trace only; no authority over mastery/publish.
Operations	outbox_events, audit_events, idempotency_records, feature_flag_audit	Append-only/retention policy; payload redaction.

7.1 Critical keys and database guards

```
UNIQUE language_profiles(user_id, language_id)
UNIQUE activity_attempts(user_id, attempt_id)
PRIMARY KEY skill_mastery_states(language_profile_id, skill, model_version)
PRIMARY KEY knowledge_mastery_states(language_profile_id, knowledge_item_id, facet, model_version)
PRIMARY KEY review_schedules(language_profile_id, knowledge_item_id, facet, scheduler_version)
INDEX review_schedules(language_profile_id, due_at) WHERE state = 'scheduled'
UNIQUE asset_versions(asset_id, version)
UNIQUE ai_prompt_templates(capability, language_scope, version)
UNIQUE ai_runs(actor_type, idempotency_key) WHERE idempotency_key IS NOT NULL
INDEX ai_runs(capability, created_at DESC)
INDEX outbox_events(status, available_at) WHERE status IN ('pending','retry')
```

ID	Вимога	Пріоритет	Критерій перевірки
TD-DATA-001	FK/transaction guard rejects cross-language attempt/mastery/review/AI session reference.	MUST	Negative DB/service fixtures.
TD-DATA-002	Critical uniqueness and referential integrity live in DB, not only UI validation.	MUST	Concurrent duplicate tests.
TD-DATA-003	AI run stores provider/model/prompt/policy version and result class without hidden reasoning.	MUST	Schema/privacy audit.
TD-DATA-004	Analytics/AI logs are never source of truth for progress/mastery.	MUST	Sink outage does not affect lesson commit.

8. Publication and content-version contract

Curriculum identity is mutable authoring metadata; learner runtime uses immutable publication manifests. Knowledge IDs survive surface wording changes so mastery is not lost. AI-generated drafts enter the same content workflow as human-authored drafts and receive no special publish shortcut.

```
Draft -> Validate -> Linguistic/Pedagogical Review -> Media/AI Provenance QA
      -> Accessibility/Technical Preflight -> Snapshot -> Publish

Rollback = atomic active-publication pointer change.
Active lesson_run remains pinned to its original publication/content versions.
```

Object	Required content
ContentVersion	entity_id, version, schema_version, payload, checksum, status, created_by/at, source_kind human|ai_assisted|imported.
Publication	course/channel/version, manifest checksum, status, previous publication, published_at.
Approval	target_content_version_id, target_checksum, gate_type, reviewer_id, decision, decided_at, notes; checksum-sensitive. Documentation baseline approval is a separate governance record.
Knowledge migration	retained/merged/split/deprecated mapping; mastery impact preview.
AI provenance	ai_run_id/prompt/model/policy refs for generated draft sections; no hidden CoT persisted.
Release package	git commit, schemas, DB migrations, content/asset manifests, validation report, approvals, feature flags.

ID	Вимога	Пріоритет	Критерій перевірки
TD-CONT-001	Published content/asset versions are immutable; edit creates new version.	MUST	Checksum/DB tests block in-place overwrite.
TD-CONT-002	Publish preflight validates exact checksums and invalidates when protected content changes.	MUST	Mutation-after-approval workflow test.
TD-CONT-003	AI draft cannot transition directly to published.	MUST	Service role/policy rejects direct publish.
TD-CONT-004	Knowledge split/merge/deprecate includes migration map and mastery impact report.	MUST	No silent skill/facet mastery loss.

9. Learning, mastery, review and outbox DDL contract

9.1 Attempts and evidence

```
CREATE DOMAIN skill_dimension AS text CHECK (VALUE IN
  ('vocabulary','grammar','listening','speaking','writing','pronunciation'));

CREATE DOMAIN mastery_facet AS text CHECK (VALUE IN
  ('recognition','meaning','recall_l2_to_l1','recall_l1_to_l2','production','spelling','orthography',
   'listening_recognition','listening_gist','listening_detail','pronunciation_target','intelligibility',
   'context_use','collocation','grammar_form','word_order','morphology','register','variant_awareness',
   'interaction_repair','interaction_goal'));

CREATE TABLE activity_attempts (
  id uuid PRIMARY KEY, user_id uuid NOT NULL, attempt_id uuid NOT NULL,
  language_profile_id uuid NOT NULL, language_id uuid NOT NULL,
  lesson_run_id uuid NOT NULL, item_id uuid NOT NULL, content_version_id uuid NOT NULL,
  client_sequence bigint NOT NULL, request_hash bytea NOT NULL,
  response_payload jsonb NULL, response_hash bytea NULL,
  outcome text NOT NULL CHECK (outcome IN
    ('correct','incorrect','assisted','accepted','speech_correct','speech_incorrect','speech_unavailable','skipped')),
  assistance jsonb NOT NULL DEFAULT '{}', feedback_code text NULL,
  evaluator_version text NOT NULL, answer_policy_version text NOT NULL,
  created_at timestamptz NOT NULL DEFAULT now(),
  UNIQUE(user_id, attempt_id),
  UNIQUE(id, user_id, language_profile_id, language_id),
  FOREIGN KEY(language_profile_id,user_id,language_id) REFERENCES language_profiles(id,user_id,language_id),
  FOREIGN KEY(lesson_run_id,user_id,language_profile_id,language_id) REFERENCES
lesson_runs(id,user_id,language_profile_id,language_id),
  FOREIGN KEY(item_id,language_id) REFERENCES activity_items(id,language_id),
  FOREIGN KEY(content_version_id,language_id) REFERENCES content_versions(id,language_id)
);

CREATE TABLE evidence_events (
  id uuid PRIMARY KEY, attempt_db_id uuid NOT NULL, user_id uuid NOT NULL,
  language_profile_id uuid NOT NULL, language_id uuid NOT NULL, knowledge_item_id uuid NULL,
  skill skill_dimension NULL, facet mastery_facet NULL, result_class text NOT NULL,
  independence text NOT NULL, difficulty numeric(6,5) NULL, evidence_weight numeric(6,5) NOT NULL,
  model_version text NOT NULL, evaluator_version text NOT NULL, content_version_id uuid NOT NULL,
  occurred_at timestamptz NOT NULL DEFAULT now(),
  CHECK (skill IS NOT NULL OR facet IS NOT NULL),
  CHECK (facet IS NULL OR knowledge_item_id IS NOT NULL),
  FOREIGN KEY(attempt_db_id,user_id,language_profile_id,language_id)
    REFERENCES activity_attempts(id,user_id,language_profile_id,language_id),
  FOREIGN KEY(language_profile_id,user_id,language_id) REFERENCES language_profiles(id,user_id,language_id),
  FOREIGN KEY(knowledge_item_id,language_id) REFERENCES knowledge_items(id,language_id),
  FOREIGN KEY(content_version_id,language_id) REFERENCES content_versions(id,language_id)
);
```

9.2 Two-level mastery and review

```
CREATE TABLE skill_mastery_states (
  language_profile_id uuid NOT NULL, skill skill_dimension NOT NULL, model_version text NOT NULL,
  estimate numeric(6,5) NOT NULL, uncertainty numeric(6,5) NOT NULL, evidence_count int NOT NULL,
  lapse_count int NOT NULL, last_evidence_at timestamptz NULL, updated_at timestamptz NOT NULL,
  PRIMARY KEY(language_profile_id, skill, model_version)
);

CREATE TABLE knowledge_mastery_states (
  language_profile_id uuid NOT NULL, knowledge_item_id uuid NOT NULL,
  facet mastery_facet NOT NULL, model_version text NOT NULL,
  estimate numeric(6,5) NOT NULL, uncertainty numeric(6,5) NOT NULL, evidence_count int NOT NULL,
  lapse_count int NOT NULL, last_evidence_at timestamptz NULL, updated_at timestamptz NOT NULL,
```

```
  PRIMARY KEY(language_profile_id, knowledge_item_id, facet, model_version)
);

CREATE TABLE review_schedules (
  language_profile_id uuid NOT NULL, knowledge_item_id uuid NOT NULL, facet mastery_facet NOT NULL,
  skill skill_dimension NULL, scheduler_version text NOT NULL, state text NOT NULL,
  difficulty numeric(10,5) NOT NULL, stability_days numeric(12,5) NOT NULL,
  due_at timestamptz NULL, lapses int NOT NULL, reps int NOT NULL, updated_at timestamptz NOT NULL,
  PRIMARY KEY(language_profile_id, knowledge_item_id, facet, scheduler_version)
);
```

Mastery invariant
Global six-skill values are derived projections from validated evidence. Review scheduling operates primarily at knowledge-item + facet level. Reveal, AI explanation, provider failure and mere exposure cannot be treated as independent productive mastery.

10. AI domain model and persistence contract

Entity	Required fields / purpose	Retention / authority
AIProviderProfile	provider_id, adapter_type, status, capabilities, data_policy_class, regions, training/retention flags, free_status, verified_at, quota metadata.	Configuration; no learner authority.
AIModelProfile	model_id, provider_id, capability tags, languages, context/tool/structured support, cost/free metadata, status/version.	Versioned configuration.
AIPolicy	capability, actor role, data class, allowed providers/models, free_only, paid_budget, retention, fallback, version.	Server-authoritative routing permission.
AIPromptTemplate	capability, language/CEFR scope, template/version, output_schema_code, grounding policy, safety policy, status.	Production artifact; reviewed/versioned.
AIRun	actor/session, capability, provider/model/prompt/policy versions, input class/hash, result class, timing, usage/cost, validation state.	Audit/quality trace; no hidden reasoning.
AISession	learner conversation scenario, language, goal, state, bounded summary, retention expiry, policy refs.	Optional; owner/language isolated.
AIMessage	session turn metadata, role, final text or redacted reference, safety/output version.	Purpose-limited; no privileged tool authority.
AIUsageBucket	provider/model/capability/cohort/day usage, quotas, estimated cost.	Cost/rate source; not learning source of truth.
AIToolAction	ai_run, proposed tool, validated args hash, actor, policy decision, execution result.	Every action audited; privileged classes prohibited for AI Auto.

```
CREATE TABLE ai_policies (
  id uuid PRIMARY KEY, capability text NOT NULL, actor_role text NOT NULL, data_class text NOT NULL,
  mode text NOT NULL CHECK (mode IN ('disabled','manual','assist','auto')),
  free_only boolean NOT NULL DEFAULT true, paid_budget_microunits bigint NOT NULL DEFAULT 0,
  allowed_provider_ids uuid[] NOT NULL DEFAULT '{}', allowed_model_ids uuid[] NOT NULL DEFAULT '{}',
  retention_policy_id uuid NULL, fallback_policy jsonb NOT NULL, version text NOT NULL, status text NOT NULL
);

CREATE TABLE ai_runs (
  id uuid PRIMARY KEY, actor_type text NOT NULL, actor_id uuid NULL, language_profile_id uuid NULL, language_id uuid NULL,
  capability text NOT NULL, provider_id uuid NULL, model_id uuid NULL,
  prompt_template_id uuid NULL, prompt_version text NULL, policy_version text NOT NULL,
  input_class text NOT NULL, input_hash bytea NULL, result_class text NOT NULL,
  output_ref uuid NULL, validation_state text NOT NULL, latency_ms int NULL,
  input_units bigint NULL, output_units bigint NULL, estimated_cost_microunits bigint NOT NULL DEFAULT 0,
  idempotency_key text NULL, created_at timestamptz NOT NULL DEFAULT now(),
  CHECK (actor_type <> 'learner' OR (actor_id IS NOT NULL AND language_profile_id IS NOT NULL AND language_id IS NOT NULL)),
  FOREIGN KEY(language_profile_id,actor_id,language_id) REFERENCES language_profiles(id,user_id,language_id)
);
```

ID	Вимога	Пріоритет	Критерій перевірки
TD-AIDATA-001	AIRun stores final structured trace, not hidden chain-of-thought.	MUST	Schema/log review finds no reasoning field.
TD-AIDATA-002	Sensitive data class blocks provider not approved for that class even if free/cheaper.	MUST	Routing negative test.
TD-AIDATA-003	AI usage/cost ledger is checked before provider call.	MUST	Zero-budget test makes no billable request.
TD-AIDATA-004	AI session is language/owner isolated and deletable according to policy.	MUST	IDOR + deletion lifecycle tests.

ЧАСТИНА C. CONTENT, EVALUATION ТА LEARNING RUNTIME

11. Authoring tree, release package and localization

```
/content/english/a1/module-01/lesson-01.yaml
/content/english/a1/module-01/recap.yaml
/content/czech/a1/module-01/lesson-01.yaml
/content/czech/a1/module-01/recap.yaml
/content/shared/locales/uk-UA/*.yaml
/content/schemas /content/fixtures /content/migrations
/assets/manifests /assets/specs
/ai/prompts /ai/policies /ai/evals
```

- Files/CMS drafts are authoring source; production runtime reads validated PostgreSQL publication projection and concrete asset versions.
- English/Czech trees validate against same schema and renderer registry; language-specific rules live in evaluator/strategy metadata.
- AI prompt/policy/eval artifacts are versioned source code/content artifacts and pass review before production use.
- No PDF import dependency in v1; no live AI generation required to open a standard activity.

Release artifact	Required content
content-release.json	release id, schema set, git commit, source checksums, generated_at
publication-manifest.json	course/channel/version, entity/content/translation/asset versions, checksum
validation-report.json	schema, refs, language, knowledge/facet mapping, evaluator, media, a11y, originality, AI provenance gates
ai-release.json	enabled capabilities, policy/prompt/model allowlist versions, free_only/budget, eval report, fallback config
migration-report.json	old/new DB/content/AI schema fixtures, ID mappings, conflicts, rollback class
approval-record	reviewer/gate/target checksum/result/notes/date; never inferred from file existence

12. Lesson Content Schema v1.7


```
{
  "schema_version": "1.7",
  "entity_id": "0198f0a0-1111-7abc-8def-111111111111",
  "stable_code": "E-A1-M01-L01",
  "content_version_id": "0198f0a0-2222-7abc-8def-222222222222",
  "version_code": "6",
  "language": "en", "variety_policy": "en-GB-default_en-US-aware",
  "cefr": "A1",
  "activities": [{
    "id": "0198f0a0-3333-7abc-8def-333333333333",
    "stable_code": "E-A1-M01-L01-A07",
    "type": "listen_repeat",
    "completion_policy": "speech_binary_nonblocking_v1",
    "knowledge_refs": [{"knowledge_item_id":"0198f0a0-4444-
7abc-8def-444444444444","stable_code":"ENG_K_PRON_H","role":"practice"}],
    "evidence_map": [{"skill":"pronunciation","facet":"pronunciation_target","weight":0.6}],
    "media_refs": [{"asset_id":"0198f0a0-5555-
7abc-8def-555555555555","asset_stable_code":"AUD_ENG_HELLO","asset_version":2,"role":"model"}]
  }]
}
```


- `ai_extensions` is optional. Its absence, disabled policy, quota exhaustion or provider outage cannot invalidate the activity or lesson.
- Generated practice is ephemeral/practice-only unless converted into a reviewed versioned content item through normal CMS workflow.
- AI hint must respect deterministic hint/reveal policy and cannot reveal answer earlier than activity policy permits.
- Content schema never stores provider-specific request/response objects.

13. Activity registry and type contracts

Family	Types	Evaluator/completion
Intro	scene_intro, vocabulary_intro, sentence_intro	Exposure; no mastery from view alone.
Selection	multiple_choice, image_choice, classification	Deterministic accepted answer; correct-or-reveal.
Mapping	matching, pair_match	Resolve set; ambiguity validation.
Construction	gap_fill, word_builder, sentence_order	Language-aware form/orthography policy.
Production	full_typing, translation, writing	Structured evaluator/rubric; facets explicit.
Receptive	listening_gist/detail, reading, dictation	Transcript policy + accessibility override metadata.
Dialogue	dialogue_completion, dialogue_chain	Persistent turn state; speaker/avatar/voice binding.
Speech	listen_repeat, read_aloud, role_play	Binary usable feedback; neutral unavailable; no unrelated mastery.
Grammar	interactive_grammar, contrast, error_correction	Staged reveal; rule built through examples.
Assessment	integrated_task, module_recap, optional_level_test	Coverage/evidence, not automatic certificate.
AI optional	ai_explain, ai_hint, ai_practice, ai_writing_feedback, ai_conversation	Capability call, grounded output, non-AI fallback; cannot replace core evaluator.

ID	Вимога	Пріоритет	Критерій перевірки
TD-ACT-001	Registry maps type+schema major to renderer/evaluator/fixtures.	MUST	Every published type has English+Czech fixture.
TD-ACT-002	AI activity capability is registered separately from deterministic activity type.	MUST	Provider disable does not break renderer.
TD-ACT-003	Unsupported schema/version yields recoverable state, not blank screen/corrupt attempt.	MUST	E2E invalid fixture.
TD-ACT-004	Randomized presentation stores seed/order sufficient for debugging.	SHOULD	Attempt replay reconstructs presentation.

14. State machines and completion contracts

14.1 Scored activity

```
ready -> active -> submitted -> correct | incorrect
incorrect -> feedback -> retry | reveal
reveal -> assisted -> completed
correct -> completed
completed -> evidence_recorded -> scheduled_review

Auto-advance allowed only after authoritative correct or explicit reveal/assisted completion.
```

14.2 Pronunciation binary UX with safe fallback

```
idle -> permission_request -> recording(pulsing) -> processing
processing -> speech_correct | speech_incorrect | speech_unavailable

speech_correct     UI: correct / "Далі" or auto-advance
speech_incorrect   UI: try again + short target-specific cue; retry/continue per activity policy
speech_unavailable UI: neutral unavailable; retry/model audio/continue

Denied mic, no mic, timeout, low confidence, provider failure -> speech_unavailable.
No negative mastery from speech_unavailable; no raw provider score shown as truth.
```

Pronunciation rule
Binary learner-facing feedback is permitted only when the speech result is usable and sufficiently confident under a versioned language/target-feature policy. It is not a CEFR certificate or accent imitation score. A mismatch may affect only the targeted Pronunciation facet, never Vocabulary/Grammar meaning by default.

14.3 AI conversation/help states

```
AI help: idle -> policy_check -> grounded_request -> waiting -> result_ready | ai_unavailable
AI conversation: active -> learner_turn -> policy_check -> waiting_model -> ai_turn | fallback_scripted -> ... -> ended

ai_unavailable covers disabled feature, quota, 429, provider outage, blocked data route, invalid output.
Fallback returns deterministic explanation/hint/scripted practice; no lesson dead end.
```

ID	Вимога	Пріоритет	Критерій перевірки
TD-STATE-001	Only one primary state controls each activity transition.	MUST	State/property tests.
TD-STATE-002	Speech and AI unavailable states are explicit neutral states, not errors converted into success/failure.	MUST	Fault matrix.
TD-STATE-003	AI output cannot directly set correct/mastery/completed.	MUST	Tampered/AI response tests.
TD-STATE-004	Reduced-motion removes continuous mic pulse while preserving textual/static recording state.	MUST	A11y motion test.

15. Answer evaluation and normalization

Dimension	English	Czech	Invariant
Unicode	NFC	NFC	Canonical equivalence only.
Case	Locale-aware except objective proper name	Locale-aware except objective proper name	Original retained; no language repair.
Whitespace	Trim/collapse where policy allows	Same	No word-order change.
Punctuation	Usually tolerant; objective-specific strictness	Same	Evaluator versioned.
Contractions	Explicit accepted variant sets	n/a	Meaning/form evidence separated.
Diacritics	n/a	Strict in form tasks; never globally stripped	Meaning may be separate from spelling facet.
Register	Usage note where relevant	ty/vy + formal/informal variants	Register mismatch visible.
Morphology	be/article patterns	být, inflected place/forms	Wrong form never normalized to correct.
Speech	Selected variety + target feature + policy confidence	cs-CZ + target feature + policy confidence	Usable -> correct/incorrect; unusable -> speech_unavailable.
AI free response	Validated rubric/schema only	Validated rubric/schema only	Advisory by default; no direct mastery unless explicit validated evidence gate.

16. Attempt command, transaction and idempotency

```
Client queue -> API validation -> deterministic evaluator -> DB transaction -> outbox -> async consumers

Inside one transaction:
  attempt + evidence + lesson position + mistake updates + mastery/review projection commands + outbox event

Outside transaction:
  analytics, optional AI help, media generation, provider callbacks

Speech provider processing, if needed, finishes before authoritative speech outcome commit or returns speech_unavailable.
No provider network wait occurs while core DB transaction is open.
```

ID	Вимога	Пріоритет	Критерій перевірки
TD-ATT-001	UNIQUE(user_id, attempt_id) gives exactly-once credit under at-least-once delivery.	MUST	Concurrent retry test creates one effect set.
TD-ATT-002	Attempt/evidence/lesson state/outbox commit atomically.	MUST	Injected failure leaves all-or-none.
TD-ATT-003	Accepted attempt references exact content/evaluator/mastery/scheduler versions.	MUST	Historical replay explainable.
TD-ATT-004	Provider call never holds core DB transaction.	MUST	Trace review + timeout fixture.

17. Evidence and two-level mastery model

Evidence class	Skill projection	Facet target	Rule
Meaning choice	Vocabulary	meaning/recognition	Positive if unassisted; reveal gives no independent evidence.
L1->L2 typing	Vocabulary/Writing	production + spelling	Higher productive weight than recognition.
Grammar gap	Grammar	grammar_form/word_order/morphology	Depends on objective; do not update unrelated meaning.
Listening gist/detail	Listening	listening_gist / listening_detail	Transcript support reduces evidence quality.
Role-play typed	Writing; optional Grammar only when the typed response explicitly assesses form/register	Writing: production/context_use/register; optional Grammar: grammar_form/register/morphology as declared	Never creates Speaking or Pronunciation evidence without usable learner audio. Additional Grammar evidence is allowed only when evidence_map explicitly declares a deterministic/rubric-scored grammar/register target.
Usable spoken turn	Speaking	production/context_use; Pronunciation separately	Speaking and pronunciation evidence emitted separately.
speech_correct	Pronunciation	evidence_map facet: pronunciation_target or intelligibility	Bounded positive evidence to the exact Pronunciation facet declared by the activity; no hardcoded facet substitution.
speech_incorrect	Pronunciation	evidence_map facet: pronunciation_target or intelligibility	Bounded practice-needed/lapse evidence to the exact declared Pronunciation facet only; never global language failure.
speech_unavailable	None	None	No mastery/scheduler update.
AI explanation/hint	None directly	None directly	Support event only; later independent attempt supplies mastery.
AI writing feedback	Writing only if validated rubric/evidence gate enabled	objective-specific facet	Default advisory; must retain rubric/model/prompt versions.

ID	Вимога	Пріоритет	Критерій перевірки
TD-MAST-001	Evidence immutable; both projection layers recomputable by model_version.	MUST	Recompute fixtures.
TD-MAST-002	Same knowledge item across lessons updates same facet state.	MUST	Duplicate surface test.
TD-MAST-003	Global skill is derived aggregation, not sole source of truth.	MUST	Reset/transition never deletes facet evidence accidentally.
TD-MAST-004	AI assistance reduces independence and cannot masquerade as clean retrieval.	MUST	AI-help then answer test.
TD-MAST-005	AI-generated result may affect mastery only through explicit validated evidence gate.	MUST	AI evidence permission test.

18. Mistake remediation, reset and Review Engine

- MistakeInstance is created for scored incorrect/form/register/spelling/pronunciation target mismatch when policy says it is a lapse; speech_unavailable creates none.
- Lesson result opens remediation for unresolved scored items. Correct/reveal can close interaction, but lapse history remains immutable and review interval stays shorter.
- Reset lesson/module changes course position and current projections according to explicit scope; historical evidence is not silently erased. Full language reset is a confirmed audited job.
- Review candidates are built inside active language profile and primarily by knowledge+facet: due -> unresolved mistakes -> weak/high uncertainty -> topic/collection -> limited recycle.
- Near-duplicate surfaces are spaced; correct-with-hint, retry, reveal and clean retrieval have different scheduling effects.
- AI may generate an extra practice surface from a bounded template/knowledge target, but scheduler ownership/due state remains deterministic server logic.

ID	Вимога	Пріоритет	Критерій перевірки
TD-REV-001	Review query begins with language_profile_id and cannot return foreign-language candidate.	MUST	Mixed dataset test.
TD-REV-002	Scheduler is deterministic/versioned for same state/result/time.	MUST	Golden tests.
TD-REV-003	AI-generated practice cannot mutate curated knowledge ownership or due date by itself.	MUST	Practice integration test.
TD-REV-004	Session cap/ETA and expiry prevent endless review or duplicate credit across devices.	SHOULD	Multi-device test.

ЧАСТИНА D. AI, MEDIA, SPEECH ТА ADMIN AUTOMATION

19. AI for Learning runtime contract

Capability	Grounding / output	Fallback / evidence
Explain differently	Current activity, approved explanation, knowledge refs, learner language/CEFR; concise structured text.	Use deterministic curated explanation. No mastery update.
Adaptive hint	Activity hint policy + current errors/support level; output schema caps hint depth.	Use stored hint/reveal ladder. Assistance is recorded.
Extra practice	Knowledge target/facet, level, accepted variants, distractor constraints; ephemeral validated activity JSON.	Use curated review item/template. Practice-only unless reviewed into content.
Writing feedback	Objective/rubric/accepted language scope + learner text minimal payload.	Static rubric/examples; advisory by default.
Conversation/role-play	Module scenario, CEFR, role, language/register, known target chunks; bounded session summary.	Scripted dialogue/role-play route.
Voice conversation	Existing STT -> AI text -> TTS pipeline behind adapters.	Text/scripted route; mic/provider failures never block core.

Grounding rule
AI is allowed to personalize presentation and practice, not invent the curriculum contract. Knowledge graph, published lesson content, accepted variants and approved rubrics are authoritative. When grounded evidence is insufficient, AI must say it cannot reliably answer or use the non-AI fallback.

ID	Вимога	Пріоритет	Критерій перевірки
TD-AILRN-001	Learner AI request has capability, language, CEFR, data class, grounding refs and output schema.	MUST	Contract tests for English+Czech.
TD-AILRN-002	AI hint cannot reveal more than deterministic hint policy allows.	MUST	Adversarial prompt/hint test.
TD-AILRN-003	AI-generated extra practice is schema/evaluator validated before presentation.	MUST	Invalid output retry/fallback.
TD-AILRN-004	Conversation session is bounded by turns/context/retention and has scripted fallback.	MUST	Quota/outage/deletion tests.
TD-AILRN-005	AI output never becomes automatic language judgment without validated rubric/evidence gate.	MUST	Mastery protection test.

20. Media, audio, voices and speech

Object	Contract
MediaRequirement	target entity/purpose, language/variety, kind, dimensions/duration, generation spec version, accessibility, required flag.
Asset	stable asset_id, purpose/language/status/provenance/rights, active version pointer; no provider URL identity.
AssetVersion	object key, checksum, MIME, dimensions/duration, codec/loudness, prompt/script hash, QA state.
Voice	language/variety, role/profile, provider adapter ref, licence/consent where relevant, version/status.
Character	versioned visual reference/style profile, linked voice, contexts; no real-person clone without lawful basis/consent.
SpeechResult	language/variety, engine/model, score/confidence/error tags, threshold policy, processing duration; raw recording normally absent.

- Normal audio required when activity marks audio required; slow uses pitch-preserving transform or reviewed slow asset.
- Microphone permission requested only after explicit action; recording control pulses while active unless reduced-motion preference is on.
- Usable/high-confidence speech -> correct or try again. Low confidence, denied mic, timeout/provider failure -> speech_unavailable with retry/model audio/continue.
- Raw microphone bytes are transient by default and absent from logs/analytics/object storage unless explicit separate policy/consent.
- Generated image/audio follows same provenance, originality, QA, accessibility and immutable versioning whether produced by AI or non-AI tooling.

21. AI orchestration, provider routing and cost controls

```
Learner/Admin capability request
  -> AI Policy Engine (role + data class + feature flag + mode + budget)
  -> Grounding Builder (minimal approved context)
  -> Provider Router (capability + language + privacy + quota + cost)
  -> Provider Adapter (local/free/paid optional)
  -> Structured Output Validator + safety checks
  -> result / fallback
  -> AIRun + usage + audit events

No provider selected -> fallback, not hidden paid escalation.
```

Control	Design
Provider classes	Local model; approved free cloud; optional paid provider only after explicit enable/budget.
Routing priority	Privacy/data eligibility -> capability/language -> free_only policy -> remaining quota -> quality/latency -> configured priority.
Paid gate	Hard pre-call check: provider/model billable + paid_enabled=false or budget<=0 => blocked before network request.
Fallback	Alternative approved local/free provider, deterministic core path, or typed ai_unavailable.
Quota	Per capability/user/cohort/provider/model/day; concurrency limits and circuit breaker.
Prompt governance	Template/version/output schema/grounding/safety policy pinned in AIRun.
Data minimization	Prompt gets only necessary context; raw secrets/credentials never included; sensitive data only to approved provider class.
Telemetry	Latency, 429/5xx, fallback, usage units, estimated cost, validation result; no hidden reasoning.

ID	Вимога	Пріоритет	Критерій перевірки
TD-AI-001	Provider routing is policy-aware and cannot bypass data classification for cheaper/free provider.	MUST	Sensitive-route negative tests.
TD-AI-002	Zero paid budget prevents all billable provider calls.	MUST	Mock billing route sees zero network calls.
TD-AI-003	Provider/model can be disabled by feature flag/emergency switch without redeploy.	MUST	Fault/incident drill.
TD-AI-004	Structured output failure triggers bounded retry then fallback; malformed output is never trusted.	MUST	Invalid JSON/schema fixtures.
TD-AI-005	Provider SDK/types do not leak into domain, content or learner API.	MUST	Adapter substitution compile/contract test.

22. AI Administration: Manual / AI Assist / AI Auto

Action	Manual	AI Assist	AI Auto v1
Draft lesson/exercise/explanation	Human creates	AI produces proposal/diff	Allowed for draft only when capability allowlisted
QA/anomaly issue	Human review	AI suggests issue/cause	May create low-risk issue automatically
Low-risk metadata fix	Human edit	AI proposal	Allowed with schema validation + diff + audit
Media prompt/script	Human author	AI proposal/job prep	Allowed subject to media gates/quota
Generate media job	Human starts	AI prepares parameters	Allowed only for approved non-sensitive requirement class
Publish / rollback	Human admin	AI may summarize preflight	PROHIBITED
Delete/disable account	Human authorized	AI may explain scope	PROHIBITED
Change roles/MFA/security	Human authorized	AI may recommend	PROHIBITED
Change paid budget/provider policy	Human authorized	AI may recommend	PROHIBITED

- Every AI Auto operation has capability ID, risk class, allowed target state, actor/service identity, policy version, before/after diff and audit event.
- AI Auto writes only to draft/non-authoritative data or schedules approved low-risk jobs. Published content/mastery/account/security state remain human/server-controlled.
- Admin UI has emergency disable for all AI, learner AI and admin AI separately, plus provider-specific disable.
- AI recommendations do not count as approval. Linguistic/pedagogical/technical/media gates retain human or separately validated authority.

ID	Вимога	Пріоритет	Критерій перевірки
TD-AIADM-001	AI Auto uses server-side allowlist by capability/risk/target state.	MUST	Negative permission matrix.
TD-AIADM-002	Privileged/destructive actions are Manual-only in v1.	MUST	Prompt injection/tool call cannot execute them.
TD-AIADM-003	AI draft/proposal stores provenance and diff before acceptance.	MUST	Admin workflow audit fixture.
TD-AIADM-004	AI mode/settings changes themselves require human admin + audit.	MUST	RBAC/MFA setting test.

23. Jobs, queues, outbox and callbacks

Job class	Examples	Behavior
Content	validate, coverage, manifest, publish preflight	Deterministic input checksum; no publish without approval.
Media	image/audio generation, transcode, QA	Quota, provider timeout, immutable ingest, bounded retry.
Speech	async scoring if request budget exceeded	Transient encrypted/direct bytes; neutral fallback on failure.
AI learner	practice generation, conversation turn, writing feedback	Policy/budget/grounding snapshot, bounded context, output validation.
AI admin	draft/QA/proposal/metadata/media orchestration	Dry-run/diff/audit; no privileged direct mutation.
Privacy	export, deletion, retention cleanup	High priority/resumable; downstream status.
Notifications	reminder/email/push	Consent/timezone/quiet hours at send time.
Projection	analytics/search/coverage	Rebuildable; poison event isolated; core commit unaffected.

```
queued -> running -> succeeded
       |          \-> retry_wait -> running
       \-> failed_terminal | cancelled | dead_letter

job_id, job_type, deterministic_key, input_ref/hash, attempt_count, max_attempts,
available_at, lease_until, timeout_ms, error_code_safe, provider/model_version, usage/cost
```

ID	Вимога	Пріоритет	Критерій перевірки
TD-JOB-001	Domain transaction writes outbox event atomically; consumers idempotent.	MUST	Crash/double-delivery tests.
TD-JOB-002	Provider callbacks verify signature/origin/timestamp/job/provider/model/replay window.	MUST	Forged/replay callbacks rejected.
TD-JOB-003	Retries bounded with backoff+jitter and non-retryable classification.	MUST	Permanent error reaches terminal state.
TD-JOB-004	Queue metrics expose age/failure/retries/usage/cost by safe bounded labels.	MUST	Fault injection alerts.

ЧАСТИНА E. API, WEB/PWA ТА ADMIN UI

24. API conventions and surfaces

Concern	Contract
Base	HTTPS JSON REST /api/v1; OpenAPI 3.1.2; typed client generated/validated.
Identity	Authenticated user from session; language_profile explicit for learner resources; client user_id never authority.
Validation	Runtime schema at boundary; language lineage/ownership/version checked before use case.
Errors	problem-details compatible: type/title/status/code/detail_safe/instance/correlation_id/field_errors.
Idempotency	attempt_id for learning; Idempotency-Key for destructive admin/jobs/AI generation commands.
Concurrency	ETag/If-Match/version for settings/drafts; no last-write-wins for publication/AI policy.
Caching	Private auth responses no-store unless explicitly safe; immutable published assets/content use version/checksum.
AI	Capability endpoints return typed result/fallback + policy/provider/model versions safe for diagnostics; never provider secret/key.

24.1 Learner and AI APIs

Method/path	Purpose	Critical contract
GET /catalog?language=	Published course map	Language/profile publication version.
POST /lesson-runs	Start/resume pinned lesson	Exact manifest/content versions.
POST /lesson-runs/{id}/attempts	Evaluate/persist attempt	Idempotent, server-authoritative.
POST /review-sessions	Build facet-aware review	Language hard filter; scheduler version.
GET /mastery/summary	Derived global skills + optional facet detail	Uncertainty/coverage aware.
POST /ai/explain	Optional grounded explanation	Feature/policy/quota; curated fallback.
POST /ai/hint	Bounded hint	Cannot exceed activity hint/reveal policy.
POST /ai/practice	Generate ephemeral practice	Schema/evaluator validation; practice-only policy.
POST /ai/writing-feedback	Structured rubric feedback	Advisory unless validated evidence gate.
POST /ai/conversations	Start bounded role-play	Language/CEFR/scenario/quota/retention/AI label.
POST /api/v1/ai/conversations/{session_id}/messages	Continue turn	Owner/session/context bounds; safe output/fallback.
GET/PATCH /settings	Audio, auto-advance, AI learner opt/settings where exposed	Server source of truth; no provider secret.
POST /data-export; DELETE /account	Privacy lifecycle	Reauthentication; async audited job.
POST /speech/jobs	Create/reuse speech scoring job	Idempotency-Key + attempt_id; owner/profile/item/content pinned; returns job_id + transient upload_url.
POST /speech/jobs/{job_id}/submit	Queue uploaded audio	Owner scoped; idempotent; upload checksum/TTL validated.
GET /speech/jobs/{job_id}	Poll async result	Owner scoped; terminal outcome exactly once; no raw provider score as truth.
POST /speech/jobs/{job_id}/cancel	Cancel scoring	Idempotent; settles speech_unavailable(reason=user_cancelled); late callbacks ignored.

24.2 Admin/content/AI APIs

Capability	Commands/queries	Gate
Drafting	Create/edit/validate/preview versioned structured content	Editor role + optimistic concurrency.
Review	Assign/decide linguistic/pedagogical/technical/media	Separation policy + checksum-sensitive approval.
Media	Requirements/jobs/outputs/approve/reject/regenerate	Provenance/rights + quota + QA.
Publication	Preflight/publish/promote/rollback	All approvals + MFA/admin + idempotency.
AI settings	GET/PATCH mode, provider/model priority, free_only, quotas/budget	Admin/MFA; audited; secrets never returned.
AI proposal	Generate lesson/QA/media prompt proposal	Manual/Assist/Auto policy; draft only; diff/provenance.
AI run inspection	Status/result validation/usage/fallback trace	Safe metadata; no hidden reasoning/raw secrets.
Operations	jobs/audit/coverage/provider health/cost/content issues	Privileged; retry/cancel within policy.

ID	Вимога	Пріоритет	Критерій перевірки
TD-API-001	API never trusts client-supplied score/due_at/role/ownership/published/AI mode/budget.	MUST	Tampering matrix.
TD-API-002	AI endpoints apply data classification, policy and budget before provider call.	MUST	Provider mock sees no blocked request.
TD-API-003	Delete/export and AI admin commands expose safe status only to authorized actor.	MUST	IDOR/RBAC E2E.
TD-API-004	Breaking API/content/AI contract is versioned with migration/deprecation plan.	MUST	Contract diff gate.

25. Web/PWA and LessonPlayer lifecycle

```
AppShell
  LanguageContext + RouteErrorBoundary
    LessonPlayer(run, manifest)
      Context / Instruction / Progress
      ActivityRenderer(type, schemaVersion, payload)
      Feedback + Hint/Reveal + Audio/Speech + optional AI help
      Back / Close / Continue / Resume recovery
```

State	UI responsibility	Persist/side effect
loading	Skeleton/status; no stale answer controls	Fetch pinned run/manifest with retry.

State	UI responsibility	Persist/side effect
active	One activity; accessible media/help; AI entry optional	Persist presentation seed/local position.
submitting	Prevent duplicate UI action; permit network retry	Queue/send same attempt_id.
correct/incorrect/assisted	Announce authoritative state; specific feedback	Persist server result only.
speech_*	Pulse/static record state; correct/try again/unavailable	No raw bytes long-term; bounded evidence.
ai_waiting/result/unavailable	Non-blocking status; cancel/fallback	AIRun metadata; no fabricated progress.
remediation	Repeat unresolved scored items; explicit exit warning	Mistake/evidence server durable.
completed	Outcome/mistakes/next action; relevant skill indicators only	Server completion; analytics async.
recoverable_error	Retry/reload/exit; correlation code	Retain unsent queue; no fake progress.

- Full six-skill profile lives in Progress; lesson result need not display six scores every time.
- Auto-advance respects authoritative result and speech/AI completion policy; AI unavailable does not convert a scored answer into success.
- PWA shell/offline queue may cache approved assets, but full offline course is a separate acceptance scope.
- AI conversation/help is clearly labeled as AI interaction where required by product/legal policy and always has accessible textual status.

26. Admin/CMS frontend and AI controls

Screen	Minimum capability
Course map	Language/course/level/module hierarchy, coverage, prerequisites, gaps/duplicates.
Lesson editor	Structured schema, validation pointers, preview, diff, knowledge/facet evidence mapping.
Review queue	Linguistic/pedagogical/technical/media gates, comments, checksum change warning.
Media	Requirement/job/output/provenance/asset versions/playback/approve/reject/regenerate.
AI settings	Global/learner/admin toggles; Manual/Assist/Auto; free_only; provider/model priority; quota/budget; data-policy eligibility.
AI proposals	Side-by-side proposal/diff, provenance, validator result, accept/reject/partial apply to draft.
AI operations	Run/job status, fallback/429/error, usage/cost, prompt/model/policy versions, emergency disable.
Publication	Exact preflight, manifest diff, knowledge migration, publish/rollback confirmation.
Operations	Jobs/audit/provider health/content issues/deletion/export; no raw audio/secrets.

ID	Вимога	Пріоритет	Критерій перевірки
TD-CMS-001	CMS never edits published content in place.	MUST	Save targets draft version only.
TD-CMS-002	Preview uses same learner renderer/schema.	MUST	Renderer contract test.
TD-CMS-003	AI Auto action shows policy/risk/diff/audit and remains draft/low-risk.	MUST	Negative privileged action test.
TD-CMS-004	Paid budget/provider policy change is human-admin action, never AI Auto.	MUST	RBAC/MFA test.

ЧАСТИНА F. SECURITY, PRIVACY, RELIABILITY TA DELIVERY

27. Authentication, authorization and application security

Control	Baseline
Credentials	Email/password if enabled; Argon2id parameters benchmarked/versioned; safe recovery lifecycle.
Session	Revocable opaque server session in Secure/HttpOnly/SameSite cookie; rotation after privilege change.
RBAC	learner, content_editor, linguistic_reviewer, pedagogical_reviewer, administrator, worker/service; AI service has scoped identity.
Object auth	Every protected object checks owner/role/language scope server-side; client IDs are not authority.
Privileged access	Production admin MFA + reauthentication for publish/role/delete/export/AI budget/provider policy.
AI/tool security	Tool names/args server allowlisted; AI cannot call privileged command directly; prompt injection is untrusted input.
Provider security	Timeout/circuit-break, outbound allowlist where practical, signed callback verification, response validation.
Supply chain	Dependency/container/IaC/secret scans; lockfile; SBOM/artifact provenance where supported.

ID	Вимога	Пріоритет	Критерій перевірки
TD-SEC-001	Threat model covers identity, IDOR, content supply chain, media, prompt injection, AI tools, speech privacy, offline cache and deletion.	MUST	Approved control matrix before public launch.
TD-SEC-002	AI cannot execute publish/delete/role/security/billing tools.	MUST	Red-team injection test.
TD-SEC-003	Content/uploads/provider outputs are untrusted and validated.	MUST	Malformed/polyglot/invalid AI output fixtures.
TD-SEC-004	Incident controls include session revoke, AI/provider disable, content unpublish and audit preservation.	MUST	Tabletop/fault drill.

28. Privacy, consent and data lifecycle

Data class	Examples	Default lifecycle
Account	email, credentials, status, consent refs	Until deletion/required retention; export/correction/delete flow.
Learning	attempt metadata, evidence, mastery, review	Needed for service; minimize raw answers; deletion/anonymization policy.
Voice	transient recording, score/confidence/error tags	Raw bytes not retained by default; derived result per learning policy.
AI learner	prompt context subset, final feedback/conversation, AIRun metadata	Minimized; provider eligibility by data class; conversation retention configurable/off by default where feasible.
AI admin	draft content/spec/proposal, run/provenance	Product content retention; no personal learner data unless approved purpose.
Operational	logs/traces/IP/security events	Short bounded retention; redaction/restricted access.
Analytics	pseudonymous learning/product events	Purpose/legal basis/consent by jurisdiction; no raw audio/free response by default.

- Free-tier status is never a legal/privacy justification. Provider data-use/training/retention terms must be approved for the actual data class.
- AI prompts contain no passwords, tokens, hidden system secrets or unnecessary personal fields. Sensitive learner text may be blocked or routed only to approved provider class.
- Account deletion propagates to DB, object store, caches, scheduled jobs, analytics and supported provider-held data according to policy.
- Minor/age/jurisdiction flow is a public-launch blocker; this design supplies controls but does not invent a universal legal threshold.

29. Performance, reliability and recovery

Metric / failure	Design target / behavior
Core Web Vitals	p75 LCP <=2.5s, INP <=200ms, CLS <=0.1 on supported real-user segments.
API reads	p95 <=400ms for normal cached in-region reads.
Attempt submit	p95 <=700ms excluding external async speech/media/AI.
Learner core availability	99.5% monthly first production-stage target excluding announced maintenance.
AI latency	Capability-specific timeout; UI cancellable; fallback without holding lesson transaction.
AI/provider outage	Core unaffected; help/conversation falls back/ai_unavailable; speech -> speech_unavailable.
DB recovery	First-stage RPO <=24h, RTO <=8h; quarterly restore drill.
Queue overload	Bound concurrency/leases/backoff; privacy jobs prioritized; dead-letter inspection.
Cost surge	Per-user/provider/capability quotas + paid hard budget + emergency disable.

ID	Вимога	Пріоритет	Критерій перевірки
TD-REL-001	Accepted attempt survives crash/retry/analytics/AI outage.	MUST	Fault-injection one durable effect.
TD-REL-002	Restore recreates account, publication, attempts, skill/facet mastery/review and representative media/AI config.	MUST	Restore drill.
TD-REL-003	Provider outage cannot exhaust request/DB pools.	MUST	Circuit-break/load test.
TD-REL-004	Paid AI cost cannot exceed configured hard budget by routing fallback.	MUST	Cost fault test.

30. Observability and domain-event contract

Signal	Required context	Examples
Logs	time, level, service/env/version, correlation, safe code, actor class	auth fail, publish result, AI policy block; no raw answer/token/audio.
Traces	browser -> API -> DB/outbox -> worker/provider callback	attempt transaction, AI route, media job; sampled/redacted.

Signal	Required context	Examples
Metrics	bounded labels/histograms/counters/gauges	API latency, DB pool, queue age, provider failure, AI fallback/cost, cache.
Learning quality	versioned aggregate events	item ambiguity, hint/reveal, facet lapse, speech fallback.
AI quality	capability/provider/model/prompt/policy versions	helpfulness, invalid output, fallback, proposal rejection; no hidden CoT.
Audit	publish/rollback/role/delete/export/AI mode/budget/Auto action	actor/target/action/result/time; restricted retention.

ID	Вимога	Пріоритет	Критерій перевірки
TD-OBS-001	Correlation propagates through API -> outbox -> worker/provider callback.	MUST	End-to-end staging trace.
TD-OBS-002	AI telemetry distinguishes free vs potentially billable route and actual estimated cost.	MUST	Usage dashboard test.
TD-OBS-003	No raw secret/audio/full personal payload in logs/analytics/AI audit.	MUST	Canary redaction tests.
TD-OBS-004	Runbooks cover DB, queue, provider, AI cost anomaly, bad publication, secret compromise and deletion failure.	MUST	Owner/date/drill evidence.

31. CI/CD, migrations, evals and release gates

Stage	Required checks
Pre-merge	format/lint/typecheck, unit/property, schema/content/AI prompt lint, architecture rules, secret/dependency scan.
Integration	PostgreSQL/Redis/storage, migrations, transactions/outbox/jobs, provider adapters, AI policy/router/budget doubles.
Contract	OpenAPI, JSON Schema, event schemas, renderer/evaluator, AI structured output/adapters.
UI/E2E	English+Czech core, language switch, speech states, review, AI-off/fallback, admin permissions, offline/retry.
Visual/a11y	320px/tablet/desktop, 200% zoom, long strings, keyboard/screen reader/reduced motion, AI states.
AI evals	Bilingual grounded explanation/hint/practice/writing/conversation golden cases; no answer leakage/hallucinated curriculum.
Security/privacy	ASVS/IDOR/RBAC/CSRF/XSS/headers/secrets/uploads/prompt injection/tool abuse/data routing/retention.
Reliability/cost	crash/retry/provider outage/quota exhaustion/zero paid budget/restore/rollback.
Editorial	linguistic/pedagogical/media/originality + AI draft/provenance review per lesson/module.

- Major DB/content/event/AI policy schema changes use expand -> migrate -> contract; old/new English+Czech fixtures are retained through compatibility window.
- AI prompt/template/model-policy change is treated as a versioned production artifact change and must run targeted evals before rollout.
- External provider tests use deterministic doubles in CI and limited sandbox/free integration separately; production key is not required in CI.

ID	Вимога	Пріоритет	Критерій перевірки
TD-CICD-001	Production artifact is immutable and tied to commit, DB/content/AI schema set and active publication.	MUST	Release record reconstructs deployed state.
TD-CICD-002	AI prompt/policy/model rollout has eval evidence and safe rollback/disable.	MUST	Staging eval + flag rollback.
TD-CICD-003	Zero-paid-budget test is a release gate while paid AI disabled.	MUST	No billable mock calls.
TD-CICD-004	Reference 32-scenario suite runs after renderer/evaluator/mastery/speech/AI-adapter contract changes.	MUST	CI report.

ЧАСТИНА G. REFERENCE IMPLEMENTATION, ACCEPTANCE TA HAND-OFF

32. Reference modules as executable fixtures

E-A1-M01 Hello & first contact and C-A1-M01 Dobrý den! are executable bilingual contract fixtures, not merely sample content. They must traverse the same schemas, renderers, transaction path, two-level mastery, review, media and optional AI Gateway. Czech-specific morphology/diacritics/register may use evaluator strategies but not a separate engine fork.

Fixture area	English proof	Czech proof	Shared acceptance
Content/schema	E-A1-M01 4 lessons + recap	C-A1-M01 4 lessons + recap	Same schema v1.7/registry.
Language evaluation	contractions, en-GB/en-US	diacritics, ty/vy, morphology	Versioned rules + facet mapping.
Speech	/h/, contractions, sentence stress	stress, length, salient Czech forms	Mic pulse + usable correct/try again + neutral unavailable.
Review/mastery	skills + facets	skills + facets	No cross-language candidate; reveal/hint differ.
Media	original en assets/voices	original cs assets/voices	asset_id/version/provenance/a11y.
AI off	Curated explanation/hints/scripted practice	Same	All lessons/recaps complete without AI.
AI on	Grounded help/practice/role-play optional	Grounded Czech-aware help optional	Policy/provider independent; no direct mastery/publish.
Accessibility	keyboard/touch/reduced motion	same	No mic/audio-only dead end.

Regression baseline
Reference Module Design v1.6 defines 32 scenarios. All 32 remain mandatory regression fixtures. Technical acceptance below adds cross-cutting architecture/AI assertions and does not replace those content-level scenarios.

32.1 Technical acceptance scenarios

ID	Scenario	Expected evidence
TDA-01	New account completes English M01, switches to Czech M01, returns to English.	Independent progress/review/mastery restored exactly.
TDA-02	Replace lesson fixture content/order without renderer code edit.	Content-driven engine + recoverable unsupported schema.
TDA-03	Double tap/retry/process crash during attempt.	One authoritative attempt/evidence effect; resume works.
TDA-04	Reveal vs retry vs clean correct.	Different independence/evidence/scheduler effects.
TDA-05	Same knowledge item tested by recognition then typing.	Global skill + distinct meaning/spelling/production facets update correctly.
TDA-06	Usable speech match/mismatch.	Learner sees correct/try again; only pronunciation facet affected as policy defines.
TDA-07	Mic denial/low confidence/provider outage.	speech_unavailable, retry/continue, zero mastery penalty.
TDA-08	Required media missing/bad checksum.	Publish blocked; no runtime provider URL fallback.
TDA-09	Publish v2 while learner active on v1 then rollback.	Active run remains pinned; old evidence replayable; rollback atomic.
TDA-10	All AI disabled.	English+Czech core, review, remediation and media playback complete normally.
TDA-11	AI free provider quota exhausted.	Approved local/free fallback or ai_unavailable; no paid escalation.
TDA-12	Paid model configured but paid_enabled=false/budget=0.	Provider receives zero requests; user gets fallback.
TDA-13	Sensitive learner text with unapproved provider first in route.	Router skips/blocks provider; data not sent.
TDA-14	AI extra practice returns invalid schema or ambiguous answer.	Validator retries boundedly then fallback; item not shown unvalidated.
TDA-15	Prompt injection asks admin AI to publish/delete/change role/budget.	Policy/tool layer denies and audits; no privileged mutation.
TDA-16	AI draft accepted into content.	Stored as draft with provenance/diff; all normal publish gates still required.
TDA-17	AI conversation provider removed mid-session.	Session offers scripted fallback/clean end; no progress loss.
TDA-18	Restore from backup.	Account/publication/attempts/skill+facet mastery/review/AI config and representative media restored within RPO/RTO.
TDA-19	Evidence event attempts to reference another user's activity_attempt.	Composite owner/profile/language FK rejects the row; no evidence/mastery/review side effect is created.
TDA-20	Evidence/knowledge target uses a different language than its attempt/language_profile.	DB/service guard rejects cross-language lineage; transaction rolls back and audit/fault evidence is emitted.

33. Definition of Done

- Clean checkout builds with pinned toolchain and validated configs; CI runs migrations and bilingual fixtures.
- English and Czech use the same code path; language isolation assertions pass for progress, skill/facet mastery, review, collections, AI sessions and caches.
- Attempt path is idempotent/atomic; evidence immutable; global skill and knowledge facet projections are recomputable.
- Pronunciation implements explicit record/pulsing state, usable correct/try again and neutral speech_unavailable; raw mic retention policy verified.
- All required media has concrete asset_id+version, checksum, provenance/rights, transcript/alt data and approved QA.
- AI Layer is provider-neutral, disabled-safe, free/local-first, paid budget 0 by default, structured-output validated and fully auditable.
- AI Assist/Auto cannot publish, rollback, delete users, change roles/security or change paid budget/provider policy.
- AI-generated drafts/media prompts/practice store provenance; no hidden chain-of-thought is persisted.
- Core learner flow passes responsive/accessibility/security/privacy/performance/fault tests; all 32 reference regression scenarios and TDA scenarios pass as applicable.
- Staging publish/rollback, provider outage, deletion propagation and restore drills produce timestamped evidence.

34. Implementation sequence

Stage	Technical output	Exit gate
Legacy descriptive phase A Contracts	Approved specs/ADR template, monorepo/toolchain, DB/content/activity/AI schemas, bilingual fixtures, threat/privacy/a11y plans.	English+Czech schemas validate; AI-off and zero-paid-budget tests defined.
Legacy descriptive phase B Reference implementation	Production-oriented Web/API/DB skeleton + LessonPlayer + shared renderers + language switch; no throwaway prototype.	No dead ends; same engine; staging/preview deploy from private repo.

Stage	Technical output	Exit gate
Legacy descriptive phase C Persistence/identity	Identity, language profiles, catalog/publications, migrations, CI/CD, observability baseline.	Login + switch + pinned manifest + reload.
Legacy descriptive phase D Learning core	Attempt idempotency/transaction, immutable evidence, skill/facet mastery, mistakes, resume/offline.	English/Czech E2E + retry/fault.
Legacy descriptive phase E Review/collections	Deterministic scheduler, session builder, topic/collection review, reset semantics.	Language-isolated facet-aware queues; multi-device.
Legacy descriptive phase F Audio/speech/media	Object storage/CDN, assets, voices, normal/slow, mic/speech state, generation/QA adapters.	en-GB/en-US/cs-CZ; correct/try again/unavailable; outage non-blocking.
Legacy descriptive phase G Content/Admin	CMS, approvals, immutable publish/rollback, media manifests, coverage and migrations.	Draft-to-publish/rollback; AI cannot bypass gates.
Legacy descriptive phase H AI contracts + adapters	AI Gateway, policy/router, prompt registry, usage meter; at least one local and one approved free cloud adapter spike.	AI-off core passes; same structured request across adapters; paid calls zero.
Legacy descriptive phase I Admin AI Assist	Draft/proposal/QA/media prompt functions; Manual/Assist/Auto settings and audit.	Diff/provenance; no privileged action possible.
Legacy descriptive phase J Learner AI	Explain/hints/practice/writing feedback then bounded text conversation.	Bilingual evals, fallback, privacy, quota/cost gates.
Legacy descriptive phase K Curriculum rollout	Module-by-module English/Czech A1 then higher levels through validated pipeline.	No placeholder publication; per-module QA/coverage.
Legacy descriptive phase L Hardening/launch	Security/privacy/age decisions, SLO/alerts, backups/restore, support/incident runbooks.	All acceptance + legal/product launch gates.

Parallel content work
After stable schemas and authoring validators, course maps, knowledge inventories and media requirements may proceed in parallel. Bulk generation waits for schema/versioning/review gates. AI is not a prerequisite for starting or completing core implementation.

35. Deferred choices and fixed baseline

35.1 Deferred choices

Decision	Current default / constraint	Close before
Production hosting/region	Vendor-neutral containers/PG/S3; dev may use Northflank/R2.	Production environment
Exact Next/Nest/Prisma majors	Current stable compatible with Node 24 LTS; pin after spike.	Repository bootstrap
Auth provider	Secure server session + Argon2id default; managed auth only after data/lock-in review.	Identity implementation
Redis/queue provider	Redis-compatible adapter.	Jobs scale stage
Image/TTS/STT providers	At least one adapter where feature needed; no provider URL in content.	Media integration
AI providers/models	Local/free-first; paid disabled; per-capability/privacy evaluation.	AI adapter rollout
AI conversation retention	Minimal/off by default; define before public learner AI.	Learner AI public use
Paid AI budget	0 until human admin explicitly approves new decision.	Before any billable inference
Launch countries/minors	No public registration until legal/product approval.	Public launch
Native apps/full offline	Defer until PWA telemetry/user need.	Post-web stabilization

35.2 Fixed decisions

- Lingvessa is working product name; English and Čeština are real start target languages; uk-UA is UI/explanation locale.
- en-GB default; en-US meaningful variants/voices/notes are explicit; Czech is cs-CZ with language-specific morphology/register/diacritics policies.
- One content-driven platform, modular monolith + workers, PostgreSQL source of truth, immutable versioned content/media/evidence.
- Two-level mastery: six global skill projections + knowledge-item facets; Review operates at knowledge/facet level.
- Pronunciation: usable speech -> simple correct/try again; unavailable/low-confidence/provider failure neutral/non-blocking.
- Required media is system-produced before publish; user does not search/create course images/audio.
- AI for Learning + Administration is optional/provider-neutral; Manual/Assist/Auto; AI Auto allowlist only; no privileged/destructive auto actions.
- Paid AI disabled and budget 0 by default; free/local-first; no hidden paid fallback.
- Core learning, review, progress and standard published media work without AI.
- Programming begins with bilingual production-oriented reference implementation, not a disposable prototype.

36. Sources and change control

Type	Source / use
Internal	Lingvessa Formal Specification v1.7 - product requirements and precedence.
Internal	Lingvessa Technical Specification v1.7 - technical baseline, APIs, environments, acceptance.
Internal	Lingvessa AI Integration Specification v1.4 - AI capabilities, policies, free/local-first, permissions, evals.
Internal	Lingvessa Curriculum Blueprint A1 v1.6 - pedagogical sequence, A1 maps, mastery/review implications.
Internal	Lingvessa Reference Module Design v1.6 - concrete E-A1-M01/C-A1-M01 fixtures and 32 scenarios.
Historical	Technical Design v1.6 - immediate predecessor implementation contract; v1.7 supersedes it.
Runtime	Node.js releases - source
Database	PostgreSQL versioning - source
Schemas/API	source ; source
Security/a11y	source ; source
Observability	source

Type	Source / use
Dev hosting check	source ; source ; source

Change control
Any change to language isolation, state/completion contract, mastery/evidence mapping, AI authority/permissions, paid budget policy, content/asset immutability or privacy defaults requires controlled spec/ADR/migration/test update. Do not overwrite this v1.7 or historical prior versions.

Підсумкове рішення v1.7

Будується Lingvessa як одна multilingual content-driven learning platform з English і Čeština на спільних engines. PostgreSQL та immutable versioned evidence/publications є source of truth. Mastery дворівневий: six global skills + knowledge-item facets. Pronunciation дає simple correct/try again лише для usable speech, a unavailable/low-confidence/provider failure є neutral. AI for Learning і AI for Administration працює через provider-agnostic policy/routing layer, free/local-first, paid inference disabled with budget 0 by default, and cannot become a critical dependency or privileged authority. Development may use GitHub Private + Northflank Sandbox + Cloudflare R2, while production remains portable through containers, PostgreSQL and S3-compatible storage. Після owner approval програмування починається з production-oriented bilingual reference implementation E-A1-M01 + C-A1-M01.

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 / Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.
BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.
BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical mastery/facet registry

Machine-code rule
Only facet codes in this registry may appear in mastery_facet/evidence_map. Every (skill, facet) target MUST also be listed in the eligible-pair matrix; pair mismatch blocks authoring/build/publish validation. Human labels such as gist/detail/target feature/repair map to exact machine codes.

facet code	Meaning
recognition	Recognition of a known form/choice
meaning	Meaning comprehension
recall_l2_to_l1	Target-language -> explanation-language recall
recall_l1_to_l2	Explanation-language -> target-language recall
production	Independent target-language production
spelling	Spelling of lexical/form surface
orthography	Language-specific orthographic correctness
listening_recognition	Recognition from audio
listening_gist	Gist from audio
listening_detail	Specific detail from audio
pronunciation_target	Named pronunciation feature
intelligibility	Understandability of learner speech
context_use	Appropriate use in context
collocation	Chunk/collocation integrity
grammar_form	Grammar form
word_order	Word order
morphology	Inflection/morphology
register	Register/formality
variant_awareness	en-GB/en-US or accepted language variant awareness
interaction_repair	Repair/clarification move
interaction_goal	Goal completion in interaction

Facet-to-global-skill aggregation

Global skill	Eligible facets (only when evidence_map explicitly targets the skill)
Vocabulary	recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production, spelling, context_use, collocation, variant_awareness, register
Grammar	grammar_form, morphology, word_order, register, context_use
Listening	listening_recognition, listening_gist, listening_detail, meaning
Speaking	production, context_use, register, interaction_repair, interaction_goal, intelligibility
Writing	production, spelling, orthography, grammar_form, context_use, register
Pronunciation	pronunciation_target, intelligibility

Reading is not a seventh global mastery skill. Reading tasks may emit item-facet evidence to Vocabulary and/or Grammar according to the objective. The optional Level Test may report a separate assessment_domain=reading subscore; that subscore is assessment-only and is not persisted as skill_mastery_states.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Database-enforced English/Czech isolation

Isolation invariant
Service-layer checks are not sufficient. Learner-state tables denormalize language_id where needed so PostgreSQL composite foreign keys can prove lineage.

language_profiles: UNIQUE(id,user_id,language_id), UNIQUE(user_id,language_id).
lessons/publications/content_versions/activity_items/knowledge_items: UNIQUE(id,language_id).
lesson_runs: columns id,user_id,language_profile_id,language_id,publication_id,lesson_id; UNIQUE(id,user_id,language_profile_id,language_id); FK(language_profile_id,user_id,language_id)->language_profiles; FK(publication_id,language_id)->publications; FK(lesson_id,language_id)->lessons.
activity_attempts: columns include user_id,language_profile_id,language_id; UNIQUE(id,user_id,language_profile_id,language_id); composite FK to lesson_runs and language_profile; item/content FKs include language_id.
evidence_events: columns include attempt_db_id,user_id,language_profile_id,language_id; composite FK(attempt_db_id,user_id,language_profile_id,language_id)->activity_attempts; composite FK to language_profile; knowledge/content FKs include language_id; CHECK(facet IS NULL OR knowledge_item_id IS NOT NULL). This proves evidence owner/language is inherited from its attempt.
review_schedules/collections: composite FK to language_profile and knowledge language lineage.
ai_runs: learner actor requires actor_id + language_profile_id + language_id; nullable composite FK(language_profile_id,actor_id,language_id)->language_profiles enforces ownership when present.
Complex lineage not expressible by simple FK uses DEFERRABLE constraint trigger. Release-blocking negative fixtures cover foreign-user and cross-language attempt/evidence/AI references.

Canonical pronunciation state and asynchronous speech-job contract

State	Meaning	Completion / evidence / Auto-advance
speech_correct	Usable, sufficiently confident target-feature match	Positive bounded Pronunciation evidence to the exact facet declared by evidence_map (pronunciation_target or intelligibility). Auto-advance allowed after feedback dwell if setting ON; otherwise show “Далі”.
speech_incorrect	Usable, sufficiently confident mismatch	Bounded negative/practice-needed evidence only to the exact Pronunciation facet declared by evidence_map (pronunciation_target or intelligibility). Non-blocking. Auto-advance may proceed after feedback dwell if setting ON; otherwise show Retry + Continue. Never counts as success.
speech_unavailable	Mic denied/absent, timeout, low confidence, provider failure	No mastery update. Non-blocking. Auto-advance may proceed after neutral status dwell if setting ON; otherwise Retry + Continue.

Canonical state names are speech_correct, speech_incorrect and speech_unavailable. Legacy labels speech_feedback_correct and speech_feedback_retry are invalid.

speech_job lifecycle

created -> uploaded -> queued -> processing -> succeeded | unavailable | failed | cancelled | expired. Client contract: POST /api/v1/speech/jobs (Idempotency-Key, attempt_id, language_profile_id, lesson_run_id, item_id, content_version_id) creates or returns the same owner/profile-bound job and returns job_id + single-use upload_url; PUT upload_url uploads encrypted transient audio; POST /api/v1/speech/jobs/{job_id}/submit queues scoring; GET /api/v1/speech/jobs/{job_id} polls status/result; POST /api/v1/speech/jobs/{job_id}/cancel is idempotent and finalizes the attempt once as speech_unavailable(reason=user_cancelled). Continue is the normal LessonPlayer Next action after a terminal speech outcome and does not create another attempt. succeeded maps exactly once to speech_correct/speech_incorrect using the activity evidence_map facet; unavailable/failed/expired map exactly once to speech_unavailable after bounded retry policy. Network retries reuse the same attempt_id/idempotency key/job; learner Retry after a terminal result creates a NEW attempt_id/job. Signed internal provider callbacks are job/owner/version validated; late/duplicate callbacks after settlement are ignored and audited. Raw audio is deleted at terminal settlement/cancel or TTL <=15 minutes.

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session
POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session
Logical concept	**Physical entity/table**
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles
AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRequestRecord / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical monorepo package map

/apps/web /apps/admin /apps/api /apps/worker

/packages/domain /packages/application /packages/activity-schemas /packages/lesson-engine /packages/mastery-review /packages/ai-core /packages/provider-adapters /packages/api-client /packages/ui /packages/content-tools

/packages/provider-adapters/{ai,speech,tts,image,email,storage}/...

/content/english /content/czech /assets/specs /prisma /infra /docs/adr

There is no separate /packages/ai-adapters in the canonical baseline; AI adapters are a sub-area of /packages/provider-adapters. Provider SDK types remain inside adapter subpackages.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + shared AI foundation + Admin AI

	Assist
P9	Optional learner AI + AI Auto allowlist (reuses P8 foundation)
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/ notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume
- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значення
Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.7
Accepted exceptions	None
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Canonical machine-readable source: /packages/activity-schemas/mastery-registry.json. CI MUST validate every evidence_map against the registry and eligible skill-facet pair matrix; copied tables in specifications are checksum-checked mirrors.

Appendix E. Concrete DB isolation guard set

ALTER TABLE language_profiles ADD CONSTRAINT uq_language_profile_owner_lang UNIQUE (id,user_id,language_id);
ALTER TABLE lesson_runs ADD CONSTRAINT uq_lesson_run_owner_profile_lang UNIQUE (id,user_id,language_profile_id,language_id);
ALTER TABLE activity_items ADD CONSTRAINT uq_activity_item_lang UNIQUE (id,language_id);
ALTER TABLE knowledge_items ADD CONSTRAINT uq_knowledge_item_lang UNIQUE (id,language_id);
ALTER TABLE content_versions ADD CONSTRAINT uq_content_version_lang UNIQUE (id,language_id);
activity_attempts FK(language_profile_id,user_id,language_id)->language_profiles and FK(lesson_run_id,user_id,language_profile_id,language_id)->lesson_runs;
evidence_events FK(attempt_db_id,user_id,language_profile_id,language_id)->activity_attempts plus FK(knowledge_item_id,language_id)->knowledge_items and CHECK(facet IS NULL OR knowledge_item_id IS NOT NULL);
learner ai_runs require actor_id+language_profile_id+language_id and composite FK(language_profile_id,actor_id,language_id)->language_profiles.
Negative DB fixtures MUST prove foreign-user evidence and cross-language evidence are rejected before mastery/review projection.